

Bay of Bengal model validation

Field	Entry
Region	Bay of Bengal regional workbook (LME_034).
Catch source	Sea Around Us regional catch series, 1950–2019. Reference: 2019 landings, excluding discards. Dated 3 September 2026. Regional catch and classic-PPR tables Retained catch archive
Selected article	Guénette, S. (December 2013). An exploratory ecosystem model of the Bay of Bengal Large Marine Ecosystem. 62-page report; the later 2014 distributed edition is distinct. Original report PDF Paper folder
Other known articles	Guénette (2014), User's guide to the Ecopath with Ecosim model of the Bay of Bengal Large Marine Ecosystem . Related guide; no local copy. Karim et al. (2019), Ecopath model of the coastal fisheries ecosystem of Bangladesh: unverified legacy inventory entry; separate from Karim (2018). Dutta, Paul & Homechaudhuri (2023), Food web structure and trophic interactions of the Northern Bay of Bengal ecosystem (29 groups, West Bengal subregion); local PDF . Karim et al. (2018), Ecosystem modeling of the resettled maritime area of the Bay of Bengal, Bangladesh through well-adjusted Ecopath approach (19 groups, >90,000 km ²); local PDF .
Selected model	34_1_Bay_of_Bengal_(1978). Baseline 1978; Bay of Bengal LME extended to Maldives, northern Sumatra and adjacent high seas (6,205,051 km ²). 49 source ecological groups (+1 computational diet-import group). Selected JSON Model folder Saved coefficient workbook
Other models from the same article	No other distinct static model identified. Initial and balanced inputs are states of the same 1978 baseline. The 1978–2010 Ecosim runs and alternative industrial-effort fits are scenarios of that baseline (pp6–7, 23–24, 28–29; Appendix A6).
Selection rationale	Article – ? Model – ?

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Model extraction	Diet composition matrix (Table 17) was corrected in several ways: - Table 17 seems to omit detritus feeding for all "3" groups – as arises from comparing to the matrix for "2" groups. For these groups, diet was completed to sum to 1 by completing the detritus feeding cell. These include: 2–3 Indian mackerel (5.48%), 3 Carangids (0.95%), 3 S pelagics (0.88%), 3 SM inv (3.36%), 3 Crustaceans (40%), 3 Macrobenthos (55%), 3 Meiobenthos (100%), 3 Zooplankton (10%). - Groups 46–48 (Bigeye tuna, Yellowfin tuna, Marlins) rows in Table 17 appear to be missing, and Detritus and Import groups are mistakenly put at indices 46, 47 instead of 49, 50. Missing values (for groups 46–48 as prey) were filled with zeros. - After these corrections, normalization was applied to fix rounding errors. Othe notes: - BA not reported; auto completed through LIM
GE diagnostics	OK: Recycling in living compartments ($p_{\text{living}} < 0.7$). No negative SPPR entries across the basal-source columns; $p_{\text{living}} = 0.59$ $b = 0.17$ Detritus (49) SPPR = 1.31 Sppr calculation is balanced (inflow=outflow).
TE diagnostics	OK: Recycling in living compartments ($p_{\text{living}} < 0.7$). No negative SPPR entries across the basal-source columns; $p_{\text{living}} = 0.63$ $b = 0$ (structural convention). Detritus (49) SPPR = 0.80. Sppr calculation is balanced (inflow=outflow).
SPPR calculation	Removed groups from calculation: seabirds (GE<0.01%, EE=5%, TL=3.4, sppr_TE=3.75M, sppr_GE=21K)
Geographic fit	Approximate overlap: A $\approx$ 100% of the target LME covered by the study B $\approx$ 59% of the study covered by the target LME. Note that the study extends to Maldives and high. Geographic evidence and area calculation
Temporal fit	The static model represents 1978; retained catch series spans 1950–2019 and the validation reference is 2019. Fixed model coefficients and pooled 1978 total-catch weights assume stable food-web structure, taxon composition and fishery geography across years and catch bases.
Other	The assumed Marine fishes not identified mixture contributes 23.19% of GE, 22.49% of TE and 23.23% of With Egestion PPR. Its unknown composition remains consequential after candidate correction. Method-specific exposure
Open issues and next action	

Field	Entry
Review and reproducibility	Researcher name: Ido Carmel review date: 01/10/2026 MODEL VALIDATED

Taxon mapping and coverage

Reference: 2019 landings, excluding discards.

49 source ecological groups (+1 computational import); 315 catch-taxon labels, including 28 zero-landings labels.

Total simple-chain PPR (2019): 379,324,412.047 tonnes C.

27 zero-landings labels lack a saved simple-chain coefficient and TL. They contribute zero annual PPR; no positive-landings taxon has unknown simple-chain PPR.

Method: simple trophic chain.

[Full taxon-mapping appendix](#) | [Sources and detailed evidence](#)

Overall confidence	Taxa (n)	Catch (%)	Simple-chain PPR (%)
High	66	20.0050	24.5910
Medium	244	59.1735	67.6100
Low	0	0.0000	0.0000
Very low	5	20.8215	7.7990
Unresolved	0	0.0000	0.0000

Overall confidence is the weaker necessary group-assignment or allocation-weight confidence.

Group assignment rules

Membership evidence: Guénette (2013), A1.1 pp37–41 and A1.3 pp43–45; authoritative taxonomy and habitat records. [Detailed evidence](#)

Plain-language rule	Confidence	PPR percentage
Explicit assignment in the selected source	High	60.3323
Assumed eligible set from source definitions and regional taxonomy	Medium	28.2780
Broad reporting category approximated by compatible named and residual groups	Very low	7.7990
Unambiguous fit to a completely defined source group	High	2.5513
Supported extension from source-listed relatives	Medium	0.6593
Verified same-taxon synonym of an explicit source member	High	0.3802
Supported ecological extension to a represented guild	Medium	0.0000

Allocation weight rules

Weights and assumptions: Guénette (2013), A2.1 p46; pooled 1978 model-catch proportions and transfer assumptions. [Detailed evidence](#)

Plain-language rule	Confidence	PPR percentage
Source-model catch proportions transferred across taxon composition, years and catch bases	Medium	75.4090
Entire taxon assigned to one group	High	24.5910

Rule percentages describe each component's share of the same available simple-chain PPR.

Very low decisions

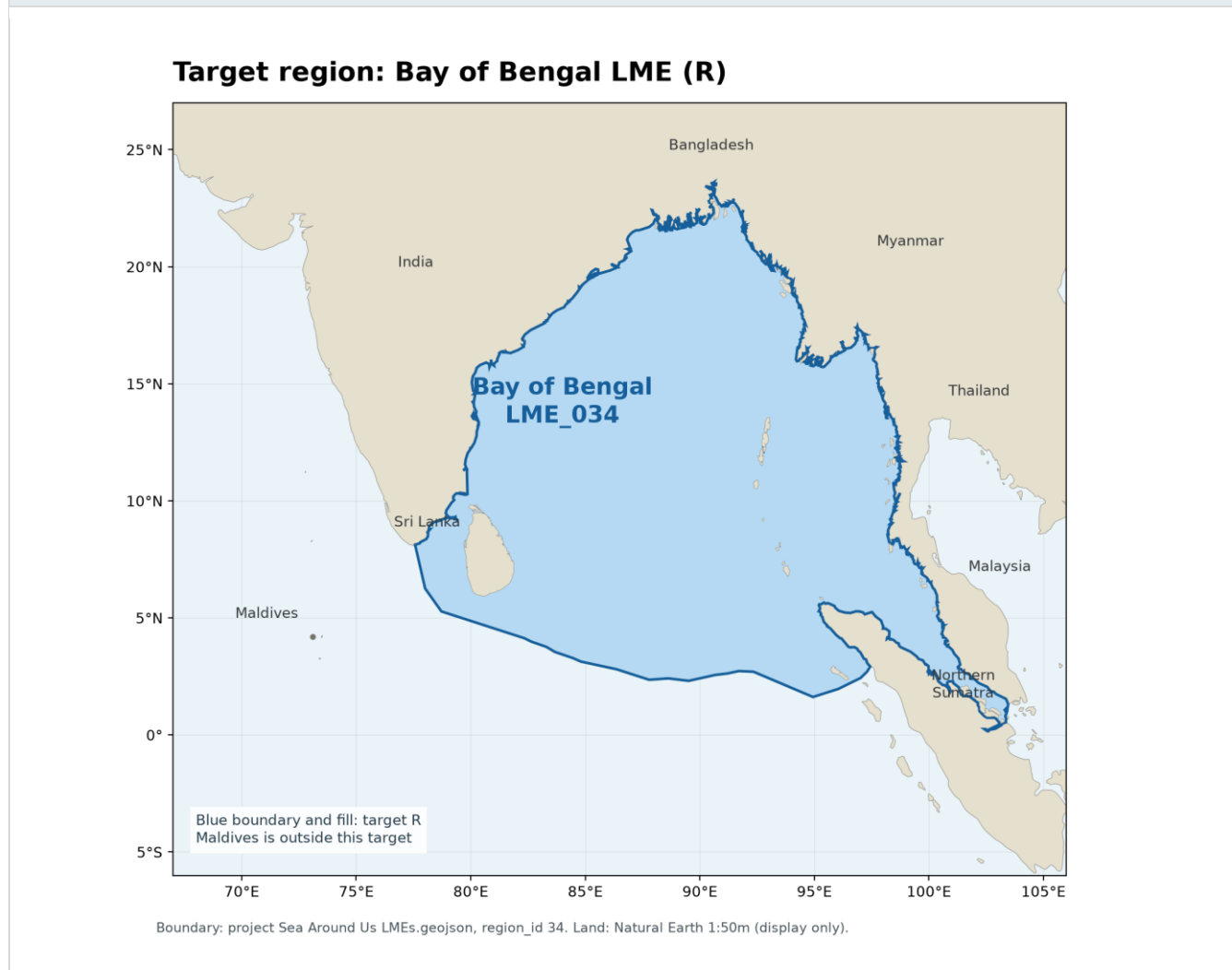
Affected taxa	Why confidence is very low
Marine fishes not identified	Unknown bony-fish composition is approximated by all 23 compatible named and residual fish pools. Separately reported tunas and billfishes do not establish their absence from unidentified catches. Source-zero S bathy remains eligible; Maldives shelf-fish pools are excluded without excluding region 1 open water. Whole-guild 1978 catches proxy unknown constituent fractions, assumed transferable across years, fisheries and catch bases.
Perciformes	The provider label is interpreted as a historical fisheries reporting concept with unknown composition. Compatible piscivore, invertebrate-feeder, carangid and small-pelagic guilds are retained; source-listed Lactarius and Atropus support the small-pelagic candidates. Eligible guilds also contain other orders, so their whole catches approximate the reporting mixture. Pooled 1978 catch proportions are assumed transferable across years and catch bases.
Marine pelagic fishes not identified	Unknown pelagic bony-fish composition is approximated by 21 compatible named and residual pools, including tunas, billfishes and source-zero S bathy. Predominantly reef/demersal commercial piscivores are excluded. Maldives shelf-fish pools are excluded without excluding region 1 open water. Whole-guild 1978 catches proxy unknown constituent fractions, assumed transferable across years, fisheries and catch bases.
Scombroidei	The provider common name and commercial class support a tunas, bonitos and billfishes reporting scope. Tuna-like, Coastal scombrids, Bigeye tuna, Yellowfin tuna and Marlins approximate the unknown mixture. Indian mackerel is excluded under the commercial-category assumption; shelf demersal pools are excluded. Pooled 1978 source-model catch proportions supply an assumed split across years and catch bases.
Malacostraca	The provider common name includes lobsters, crabs, shrimps and krill. Crustaceans, Macrobenthos and all three Zooplankton pools approximate that broad mixture, consistent with the source-listed small benthic crustaceans and euphausiids. Zooplankton candidates retain genuine zero catch and zero weight; region 1 open water remains eligible. Pooled 1978 catches supply an assumed split across years and catch bases.

Geographic evidence screenshots

Target region R and source-model study area S, used for the approximate geographic comparison.

Region area

Target region boundary



R: Bay of Bengal LME_034. WGS84 geodesic area $\approx 3,664,447 \text{ km}^2$; project Sea Around Us boundary, region_id 34. Natural Earth land is a display basemap.

Study area in the article

Study boundary for the selected model

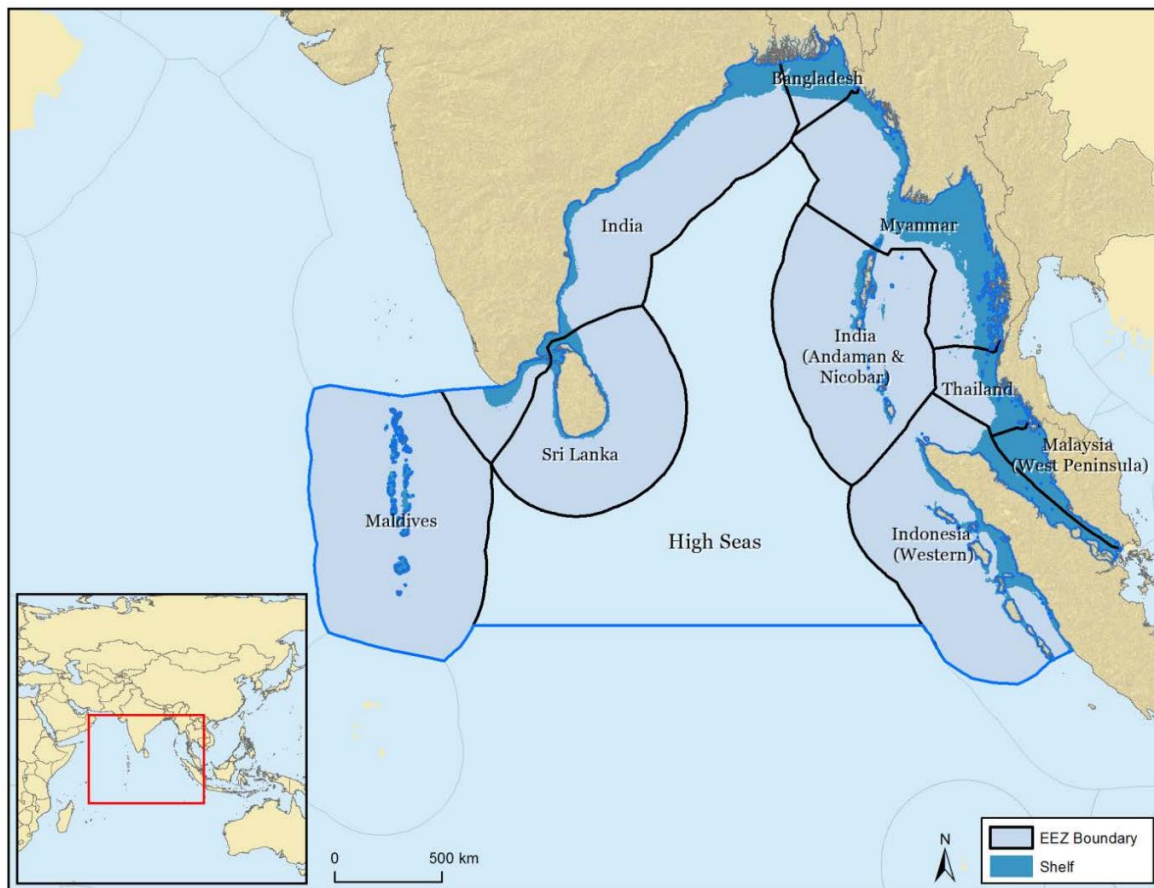


Figure 1. Map of the study area, the Bay of Bengal LME and the Maldives. Source UBC SAUP.

S: Guénette (2013), Figure 1, p8. The study includes Maldives and adjacent high seas; Table 1 gives 6,205,051 km².